ADDRESSABLE EDITING FOR RELIABLE OCR POST-CORRECTION: INTERFACES, TAILS, AND PROTOCOL COMPLIANCE

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Yifan Zhang
School of Information Science and Engineering
Lanzhou University, Lanzhou 730000, China
yfzh2023@lzu.edu.cn
https://orcid.org/0009-0002-1624-0327

ABSTRACT

Large language models can correct noisy historical OCR, but a correction that fixes a few characters can also silently rewrite a whole document. We study how the editing interface exposed to an agent changes this reliability trade-off. We implement a block-addressable document intermediate representation, typed edit application, content-addressed snapshots, and auditable session tapes with deterministic replay. On 33 matched tasks from 11 German DTA19 source documents at three synthetic noise levels, we compare eight one-shot interfaces under three model configurations. With DeepSeek V4 Flash, local block-scoped find-and-replace has median/P90/maximum cMER of 0.0084/0.0199/0.0295, while unrestricted rewriting has 0.0120/0.0389/0.9648. DeepSeek V4.1 shows the same ordering of the observed tail, with a smaller gap. MiniMax M3 frequently fails to complete the text-tool protocol, so its tool-arm scores reflect protocol adherence as well as correction quality. A global find-and-replace view with numeric keys changes V4's correction rate from 61.3% to 51.0%; because the condition also changes the prompt view, we treat this as a presentation/configuration effect rather than a pure key-only intervention. The evidence supports a conditional reliability hypothesis for protocol-compliant models, while exposing limits from small document count, one endpoint run per arm, and non-leaderboard document-level aggregation.

Keywords OCR post-correction · LLM agents · editing interfaces · reproducibility

1 Introduction

Historical collections contain systematic OCR errors, and post-correction systems must improve those errors without damaging text that was already correct. HIPE-OCRepair makes this trade-off measurable with character-level Match Error Rate (cMER), including over-correction effects that can be hidden by an average score [Ehrmann et al., 2026]. For an agent, the editing interface is part of the system: a complete replacement gives the model a document-wide destructive action, whereas a verified replacement scoped to a stable block can limit the consequences of a mistaken edit.

Prior work has shown that the interface between a language model and its computer changes software-engineering outcomes [Yang et al., 2024]; executable text-editing operations and audit traces have also been proposed for human-facing workflows [Laban et al., 2023]. We transfer this question to OCR post-correction. Our goal is an empirical characterization of interface and protocol behavior, not a universal ranking of models or tools.

We make four contributions:

- an inspectable editing harness with a typed document IR, block addressing, atomic application of each parsed operation batch, content-addressed snapshots, and replayable session tapes;

- an eight-arm ablation on matched DTA19 document tasks that separates scope, key semantics, attention instructions, and full-document output as far as the implemented prompts permit;
- a tail-aware result: local patching keeps the worst observed V4 cMER at 0.0295, while the corresponding full rewrite has one document at 0.9648, with a substantially smaller median gap;
- an explicit account of protocol compliance and presentation effects, including the M3 failure mode and the V4 correction-rate shift after adding a keyed view.

The claims are deliberately conditional. The experiments use one German development subset, one run per model/arm, and a custom document-level aggregation. The reported bootstrap intervals quantify variation over the 11 shared source documents; they do not quantify model or endpoint randomness.

2 Background and Related Work

OCR post-correction. HIPE-OCRepair evaluates historical transcripts with cMER and related measures, and identifies over-correction as a practical challenge [Ehrmann et al., 2026]. We use the repository's released scorer [HIPEval, 2026] and a matched DTA19 development subset; our document-level summaries are not official leaderboard scores.

Interfaces for language-model agents. SWE-agent treats the model-computer interface as an experimental factor [Yang et al., 2024]. InkSync and structured output formats similarly make edits executable and inspectable [Laban et al., 2023, Cheng et al., 2026]. Our setting differs in that the desired output is a minimally changed historical transcript rather than a program patch.

Reliability and compliance. Long delegated workflows can produce sparse but severe document corruption [Laban et al., 2026]. Work comparing restricted tool surfaces also finds that cost or failure behavior can change while task success remains tied [Yang et al., 2026]. These observations motivate reporting tails, effective edits, and protocol completion together.

3 Addressable Editing Harness

3.1 Document IR and operations

The input transcript is partitioned into ordered blocks. A snapshot is a sequence

$$S = ((k_1, x_1), \ldots, (k_n, x_n)),$$

where k_i is a block key and x_i its content. Patch-style tool calls are parsed into typed operations such as `find_replace(path,key,old,new)`. The parser checks the path, key, and uniqueness of `old` in the addressed scope. A valid call is lowered to IR operations and applied as one batch; a failed operation rejects that batch without a partial mutation.

The global interface checks uniqueness over the complete path. The local interface checks only the addressed block. The stable-key condition preserves keys across edits; the unstable condition uses positional identifiers and re-partitions after an edit. The full-rewrite conditions return text directly and therefore do not receive the same operation-level validation. Numeric keys are part of the model-visible view; they are not universally stripped from a rewritten output. We report leaked keys separately.

3.2 Snapshots, tapes, and replay

The harness stores block blobs and snapshot manifests by content address. Session tapes record prompts, completions, parsed calls, tool results, status, and extracted output. Replay folds the recorded calls and parser decisions without calling the model API. These facilities make post-processing changes auditable, but the full session tapes are not committed in this paper repository because the raw `runs/` directory is ignored; the manuscript instead commits a derived per-task result snapshot used by all reported figures and intervals.

3.3 One-shot protocol

Each cell uses one document and a single model turn (`max_turns=1`). Rewrite arms emit a complete transcript. Tool arms emit calls followed by `done()`. The harness supports feedback in longer sessions, but here the model has no second turn to inspect tool results or repair a parse error. Calls within a valid completion execute sequentially; atomicity applies to the IR operations of each call, not to the entire completion. A later failed call does not undo earlier successful

Table 1: Interface factors. The presentation conditions include their associated prompt clauses, so they are not all single-factor interventions.

Arm	Key view	Replacement scope	Attention clause
Global F/R	none	whole path	no
Global + keys	stable	whole path	no
Blocked F/R	stable	whole path	yes
Local patch	stable	one block	yes
Patch no-attn	stable	one block	no
Unstable local	positional	one block	yes
Rewrite	none	full output	n/a
Rewrite + keys	stable	full output	n/a

calls. The status `budget_exceeded` can mean that no completion marker was reached within the one-turn budget, rather than that the provider truncated its output. The current extracted text of such a session is retained and scored.

4 Experimental Design

4.1 Data and aggregation

We use the DTA19 German development subset distributed with HIPE-OCRepair-Bench v0.9.5. The experiment selects 11 source documents at each of three synthetic noise levels (l0, l1, l2), producing 33 matched document tasks per arm. The repository's document loader concatenates the sampled pages for each source document; the resulting score is a cMER over that concatenated transcript. In the released reports, page-level chunk alignment is not used (`chunks_aligned=0` for these document rows), so this experiment should be read as a controlled document-level study rather than an official HIPE evaluation.

We report median, P90, and maximum per-document cMER, effective edits, protocol status, and key leakage. The supplementary snapshot also contains fixed rate, broken rate, edit spans, and provider-reported tokens. Fixed rate measures the fraction of input errors corrected; broken rate measures damage relative to already-correct input characters. Edit spans count changes after scorer normalization, not byte-level edits. Key leakage counts output lines matching a numeric-prefix heuristic; it is not a percentage and can include legitimate numeric prefixes. Such prefixes are not explicitly stripped before normalized cMER scoring. The no-edit output is an anchor. The tracked file `paper/data/results.csv` contains 792 model/arm task rows plus 99 no-edit rows (one set of 33 per model).

4.2 Interface arms

The eight arms are `find-replace`, `find-replace-numkey`, `find-replace-blocked`, `patch-find-replace`, `patch-find-replace-noatt`, `patch-find-replace-unstable`, `rewrite`, and `rewrite-numkey`. The first three use global uniqueness; the patch arms use a keyed local scope. The `blocked` arm adds the two-pass attention instruction to global find-and-replace. `noatt` removes that instruction from local patching. `unstable` keeps the local view shape but changes stable keys to positional identifiers. The two rewrite arms differ only in the model-visible keyed view as implemented by the repository.

4.3 Models and uncertainty

The stored report configurations identify DeepSeek V4 Flash and V4.1 Flash on the Volcengine Responses endpoint, and `MiniMax-M3[1m]` on MiniMax's Anthropic-compatible endpoint. Each declares an output limit of 393,216 tokens; temperature and reasoning effort are unspecified and use provider defaults. The adapters' handling of reasoning and token accounting need not be identical. These are provider model identifiers, not independently verified model weights. Thus M3 changes model, endpoint, and protocol family together. There is one recorded output per model/arm/task; no sampling-replicate analysis is available.

For paired comparisons, task identifiers are aligned. We use a document-cluster bootstrap: draw 11 source documents with replacement and retain all three noise-level rows for each draw. Both arms share the same sampled indices. The committed `scripts/cluster_bootstrap.py` uses 20,000 resamples and seed 0. Mean differences average taskwise B-minus-A scores; P90 differences recompute each arm's quantile inside each resample. P90 uses sorted zero-based index $\lfloor 0.9n \rfloor$ (index 29 for 33 tasks); percentile intervals use indices $\lfloor 0.025B \rfloor$ and $\lfloor 0.975B \rfloor$ among B

Table 2: Principal results. Entries are median / P90 / maximum per-document cMER; lower is better.

Model	Local patch	Global F/R	Rewrite
DeepSeek V4 Flash	0.0084 / 0.0199 / 0.0295	0.0096 / 0.0272 / 0.0605	0.0120 / 0.0389 / 0.9648
DeepSeek V4.1 Flash	0.0078 / 0.0187 / 0.0284	0.0101 / 0.0230 / 0.0582	0.0104 / 0.0245 / 0.0605
MiniMax M3	0.0238 / 0.0605 / 0.0659	0.0232 / 0.0587 / 0.0621	0.0097 / 0.0496 / 0.7342

sorted estimates. These exploratory intervals measure source-document variation conditional on the observed outputs, not inference randomness, and are not adjusted for multiple comparisons.

5 Results

5.1 Observed tail separation

Figure 1 and Table 2 show a distributional pattern for the two DeepSeek configurations. Local patching has a lower observed maximum and P90 than full rewriting in both runs. The V4 difference is dominated by one destructive rewrite outcome; it is evidence about tail behavior in this sample, not a population-level estimate of a universal worst-case bound.

The other arms qualify this result. V4 Rewrite + keys has maximum 0.0322, close to Local patch's 0.0295, so a large tail is not inherent to every full-rewrite configuration. Conversely, V4.1 Rewrite + keys reaches 0.7483. V4.1 Patch no-attn has slightly lower P90 and maximum than Local patch. Local patch still worsens cMER relative to no edit on 6/33 V4 and 3/33 V4.1 tasks. Neither local scope nor stable identifiers guarantee harmless edits; Appendix A reports all arms including these counterexamples.

The lower patch tail is not explained by doing less editing. For V4, Local patch and Rewrite correct 65.8% and 67.4% of input errors and change 350.5 and 343.8 spans on average, respectively, while their broken rates are 0.52% and 3.86%. A document-cluster bootstrap for patch minus rewrite gives a broken-rate difference of -3.34 percentage points with 95% interval $[-9.11, -0.31]$. The same comparison for V4.1 is -0.28 percentage points $[-0.56, -0.06]$. These fixed-rate, edit-volume, and damage measures provide a mechanism-oriented complement to the cMER tail summaries; the intervals are descriptive comparisons over the 11 source documents.

The scope comparison with attention holds the stable view and attention clause fixed: Blocked F/R versus Local patch. For V4, local minus global mean cMER is -0.0072 with a document-cluster interval $[-0.0109, -0.0035]$; the P90 difference is -0.0388 with interval $[-0.0444, -0.0077]$. The corresponding V4.1 mean difference is -0.0003 with interval $[-0.0012, 0.0006]$. Without the attention clause, Global + keys versus Patch no-attn gives V4 mean difference -0.0049 with interval $[-0.0105, 0.0011]$, and V4.1 difference -0.0025 with interval $[-0.0052, -0.0003]$. Thus support depends on the model and attention condition. The keyed arms form a scope-by-instruction 2×2 comparison, but scope includes the required key argument, tool description, and matching constraints; this does not isolate an internal attention mechanism.

5.2 View presentation and attention

Changing V4 from global F/R to global F/R with numeric keys changes the correction rate from 61.3% to 51.0%, a difference of -10.3 percentage points. The cluster interval is $[-21.1, -1.7]$ percentage points. Edit spans change from 254.5 to 218.2; its interval $[-94.1, 8.2]$ includes zero. The appropriate conclusion is that this implemented view-and-prompt configuration changed behavior. The experiment does not establish that a visually non-informative number alone caused the shift.

Adding the attention clause to the global keyed arm is a separate intervention. For V4 its mean cMER difference is $+0.0012$ with interval $[-0.0048, 0.0069]$; we do not infer that the instruction alone improves accuracy. The unnumbered-to-numbered condition also changes the view description, read-result representation, and internal text/IR backend, although its intended replacement semantics stay global. V4.1's correction-rate change in that comparison is approximately -0.9 percentage points and its interval includes zero. These results do not establish a model-independent pure formatting effect.

5.3 Protocol compliance

MiniMax M3 makes the protocol boundary visible. The six tool arms complete between 1 and 15 of 33 tasks, with many cells producing no effective edit. Both rewrite arms complete all 33 tasks. Figure 2 reports completion and

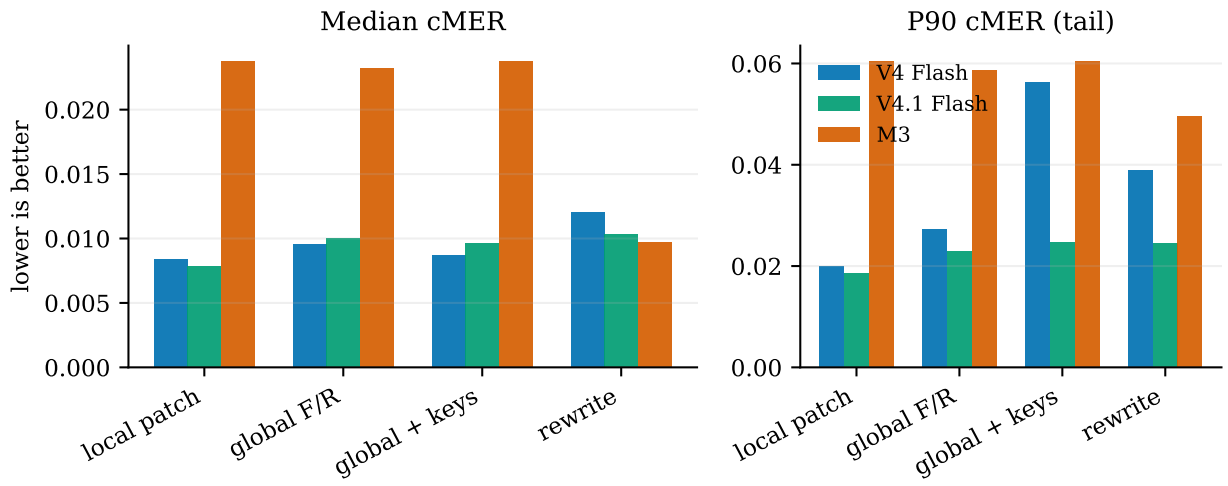


Figure 1: Median and P90 per-document cMER from the tracked result snapshot. M3 tool arms are affected by protocol non-completion.

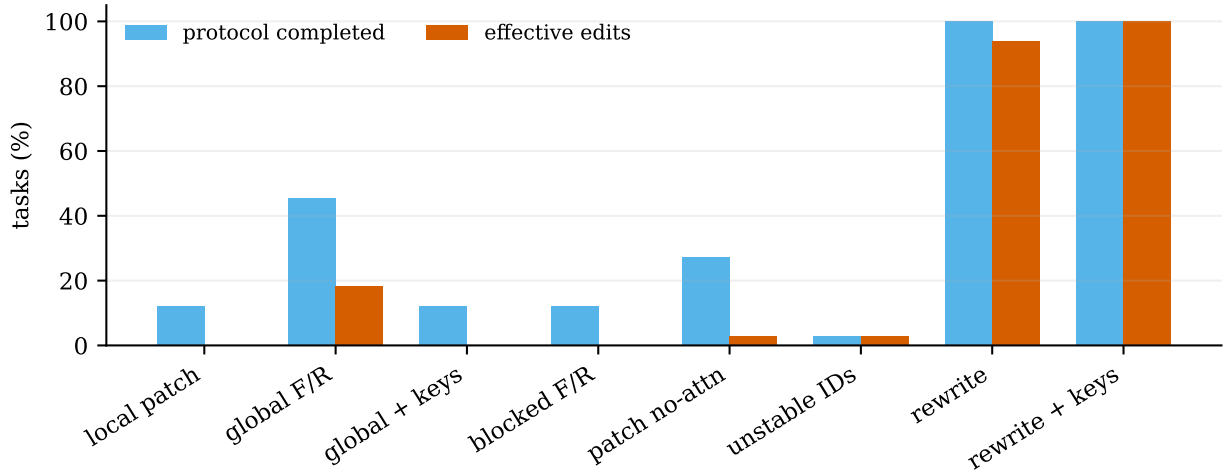


Figure 2: MiniMax M3 protocol completion and effective edits. All 33 tasks remain in each denominator. Completion does not imply a successful correction; conditioning only on completed tasks would select a different sample for each arm.

effective-edit rates. The apparent low cMER of an unedited M3 task is therefore an anchor-like outcome, not evidence that its tool interface corrected the transcript. The rewrite-with-keys arm leaks numeric prefixes in 10 of 33 documents; this is a model-view interaction and a reason to report key leakage separately.

The completion criteria differ by interface: rewrite arms do not need to emit the tool grammar or a `done()` marker. A tool completion marker can also follow rejected edits, so status alone is not a call-success metric. These counts characterize the deployed one-turn protocol and should not be read as evidence about M3's native structured-tool capabilities.

6 Discussion

The strongest supported statement is conditional: on this matched German document sample, and for the two DeepSeek configurations that mostly completed the tool protocol, a block-scoped editing interface is associated with lower observed tail cMER than global rewriting. The central scores are close enough that the result is better described as a reliability and tail observation than as a general accuracy gain.

The harness contributes an experimental control surface. A reviewer can inspect a parsed operation, verify that a failed operation batch did not partially mutate the IR, and replay deterministic post-processing. That control surface does not remove model stochasticity, endpoint drift, or prompt sensitivity.

7 Limitations and Reproducibility

The effective independent-document count is 11. The three noise levels are paired views of the same source documents, not 33 independent documents, and the bootstrap intervals reflect that structure. Each model/arm has one endpoint run, so there is no estimate of seed, sampling, or provider variance. The DTA19 German development subset is narrow; no multilingual, image-conditioned, masked-test, or human-review evaluation is included.

The custom document loader concatenates pages and the reported rows have no page-level chunk alignment. Consequently, these values are not official HIPE micro-averaged scores and should not be compared directly with leaderboard entries. The one-shot protocol also couples editing quality to instruction following; M3's incomplete tool sessions measure end-to-end protocol performance as well as editing ability.

The eight arms are not a full factorial design. Within the four stable-key tool arms, view content is held fixed while scope and the attention instruction are crossed. Outside that subset, the unnumbered-to-numbered and stable-to-positional comparisons also change descriptive instructions or execution semantics. No internal attention measurements establish how the models use the keys. Maxima and P90 from 33 correlated tasks are sensitive to individual documents; we do not estimate a worst-case bound or guarantee. A non-significant median or interval is not an equivalence result. We have no independent held-out confirmation after interface development on this dev set, no systematic native-tool or multi-turn protocol comparison for M3, and no latency, monetary-cost, or human-acceptance evaluation.

The repository provides a derived per-task snapshot, source-report hashes and model configurations, figure generator, all-arm table generator, and cluster-bootstrap implementation. This supports regeneration of descriptive statistics without API calls; it does not independently verify the predictions or their scores. Raw reports, tapes, and referenced message blobs remain outside this paper package. A complete raw artifact archive is still needed for an independent end-to-end replay audit. The document loader's newline separator is also shared by within-page text, causing the whole-document fallback even for the unchanged baseline; it should not be diagnosed as model-induced structural damage.

8 Conclusion

A language-model agent edits through an interface, and that interface can change the shape of OCR errors. In this study, local block-scoped editing reduced the observed tail for two DeepSeek configurations, while MiniMax M3 often failed before semantic editing began. The results justify reporting cMER distributions, protocol completion, and key leakage together. They also set a clear boundary on the claim: a small, single-run German document study with custom aggregation is evidence for a reliability hypothesis, not a universal guarantee.

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A Complete arm results

Tables below include every arm and the no-edit anchor. Failed completion markers remain in the quality summaries. paper/data/noise_summary.json additionally reports descriptive cMER and error decomposition by noise level; each stratum contains the same 11 sources.

Table 3: All arms for DeepSeek V4 Flash. Counts use all 33 tasks. Worse means cMER exceeds the no-edit anchor; it is not a count of catastrophic errors.

Arm	Median	P90	Max	Done	Edited	Worse	Leak
No edit	0.0238	0.0605	0.0659	33	0	0	0
Global F/R	0.0096	0.0272	0.0605	33	31	6	0
Global + keys	0.0087	0.0564	0.0659	33	28	8	0
Blocked F/R	0.0109	0.0587	0.0659	33	25	7	0
Local patch	0.0084	0.0199	0.0295	33	33	6	0
Patch no-attn	0.0089	0.0197	0.0621	33	32	8	0
Unstable local	0.0104	0.0253	0.0444	33	31	8	0
Rewrite	0.0120	0.0389	0.9648	33	33	9	0
Rewrite + keys	0.0110	0.0248	0.0322	33	33	10	0

Table 4: All arms for DeepSeek V4.1 Flash. Counts use all 33 tasks. Worse means cMER exceeds the no-edit anchor; it is not a count of catastrophic errors.

Arm	Median	P90	Max	Done	Edited	Worse	Leak
No edit	0.0238	0.0605	0.0659	33	0	0	0
Global F/R	0.0101	0.0230	0.0582	33	33	5	0
Global + keys	0.0097	0.0248	0.0479	33	32	5	0
Blocked F/R	0.0105	0.0193	0.0308	33	33	1	0
Local patch	0.0078	0.0187	0.0284	32	31	3	0
Patch no-attn	0.0094	0.0183	0.0276	33	33	5	0
Unstable local	0.0094	0.0232	0.0605	31	30	4	0
Rewrite	0.0104	0.0245	0.0605	33	33	6	0
Rewrite + keys	0.0119	0.0245	0.7483	33	33	6	1

Table 5: All arms for MiniMax M3. Counts use all 33 tasks. Worse means cMER exceeds the no-edit anchor; it is not a count of catastrophic errors.

Arm	Median	P90	Max	Done	Edited	Worse	Leak
No edit	0.0238	0.0605	0.0659	33	0	0	0
Global F/R	0.0232	0.0587	0.0621	15	6	0	0
Global + keys	0.0238	0.0605	0.0659	4	0	0	0
Blocked F/R	0.0238	0.0605	0.0659	4	0	0	0
Local patch	0.0238	0.0605	0.0659	4	0	0	0
Patch no-attn	0.0238	0.0605	0.0659	9	1	0	0
Unstable local	0.0232	0.0589	0.0659	1	1	0	0
Rewrite	0.0097	0.0496	0.7342	33	31	7	0
Rewrite + keys	0.0141	0.1167	0.7357	33	33	13	10

B Reproduction details

Run `python scripts/cluster_bootstrap.py -input paper/data/results.csv` to regenerate the reported cluster intervals and `python paper/figures/make_figures.py` to regenerate both figures. The `results.csv` columns identify model, arm, task, source document, noise level, status, cMER, fixed/broken rates, edit spans, and key leakage. The full tool prompts and implementation remain in `src/agent/numkey.py` and the corresponding TOML files under `configs/toolsets/`.